Status	Finished
Started	Sunday, 18 October 2026, 8:56 AM
Completed	Sunday, 18 October 2026, 9:00 AM
Duration	4 mins 30 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sorted()
- b. sort()
- c. order()
- d. arrange()

The correct answer is: sort()

Question **3**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It increases the number of comparisons needed
- c. It makes the search operation slower
- d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Sorting the entire list in one pass
c. Bubbling up the smallest element
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes it easier to select items based on their relationship to the rest of the items
c. It makes the data unsorted
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **8**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It allows for quicker identification of duplicates
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting helps in finding duplicates quickly
c. Sorting makes it harder to search for items
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Selection sort
c. Quick sort
d. Merge sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are split again into smaller sublists
b. They are combined to form a single sorted list
c. They are compared element by element
d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Duplicates
- b. All of the above
- c. Searching
- d. Selection

The correct answer is: All of the above

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Merging two datasets
- c. Finding a specific element in a list
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Searching for data in a list
- c. Deleting data from a list
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format